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The numerical simulation of multi-scale oil films using coupled VOF and Eulerian thin-film models

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Abstract

This paper presents a coupled ETFM-VOF framework for the numerical simulation of multi-scale thin liquid films. A depth-averaged Eulerian thin-film model (ETFM) is used to simulate the oil flow in very thin-film regions where film thicknesses are below the grid resolution while elsewhere in the domain where grid resolution is sufficient to resolve the film, a traditional Volume-of-Fluid (VOF) approach is retained. The two approaches are coupled through momentum and mass conserving source terms and a transition criterion is introduced where the total liquid volume fraction in each cell is evaluated and either the ETFM or VOF approach used depending on the sufficiency of the local grid resolution. Using this approach, thin-film flows characterised by multiple film thickness scales may be reliably simulated at a relatively lower computational cost. The model builds upon currently available ETFM and VOF approaches to thin-film modelling and represents a novel approach to the numerical simulation of multiphase flows involving a varying range of film thickness scales in space and time. A numerical test case of the 3D rimming flow inside an idealised aero-engine bearing chamber has been used to demonstrate the approach and comparisons made against high resolution VOF solutions.

Nomenclature

α_i	volume fraction of phase i
V_i	volume of phase i in control volume (c.v.)
V_T	total volume of c.v.
$h(s, t)$	film thickness
$u(s, y, t)$	film velocity
s	lateral flow direction tangential to the wall
y	flow direction normal to the wall
ρ_l	liquid density
μ_l	liquid viscosity
ρ_g	gas density

μ_g	gas viscosity
ρ_m	mixture density
μ_m	mixture viscosity
p_{gas}	core gas pressure
p_f	film hydro-dynamic pressure
σ	surface tension coefficient
κ_j	normal curvature in the j -direction
τ_w	wall shear stress
τ_i	interfacial shear stress
g_j	gravitational acceleration in the j -direction
$\alpha_{i,VOF}$	VOF domain volume fraction of phase i
$\alpha_{i,ETFM}$	ETFM domain volume fraction of phase i
$\alpha_{i,TOTAL}$	Overall volume fraction of phase i
$\alpha_{i,CRIT}$	Critical volume fraction of phase i

1. Introduction

A film may be described as a layer of liquid partially bounded by a solid substrate, with a free surface where the liquid is exposed to another fluid - usually a gas [1]. The depth of the film is typically orders of magnitude smaller than the other lateral dimensions of the liquid expanse. To distinguish thin-film from thick-film flows, a thin-film flow may be explicitly categorised as one in which the velocity field is approximately planar (two-dimensional) with the film vertical velocity component (locally normal to the solid substrate) being negligible compared to the lateral velocity components (locally parallel to the solid substrate).

Aero-engines and gas-turbine oil systems involve liquid film flows of varying scale ranging from thin-film flows to much thicker film flows interacting within the same system as shown in Figure 1. Oil is injected into the bearings to provide bearing lubrication as well as extract heat from the bearing chamber. The oil is shed from the bearing and atomised into droplets/filaments which are transported by the core air flow and

deposited onto the stationary outer wall, forming a film. The oil in the film on the stationary outer wall is then circulated around the chamber by an interfacial shear stress at the air-oil interface. In addition, gravitational, pressure gradient and surface tension forces as well as momentum transfer through droplet impact all have an influence on the film dynamics. Excess oil is extracted at a controlled rate in the sump region and recycled after undergoing a treatment process.

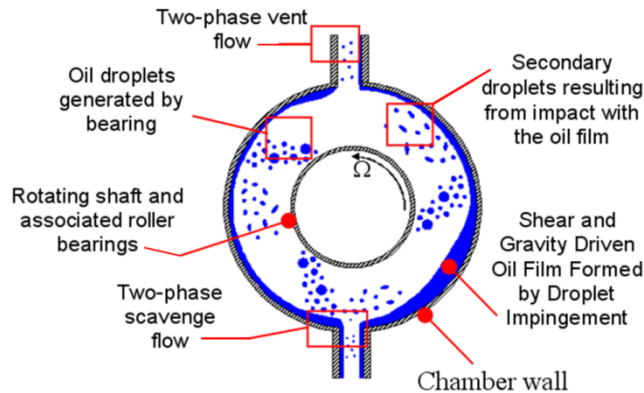


Figure 1: Schematic of a typical aeroengine bearing chamber operation

By controlling the oil injection and sump off-take rates, as well as the flow behaviour of the film, optimal heat transfer and oil quality (in terms of degradation of lubrication properties) can be achieved. It is therefore necessary to provide reliable numerical tools for design engineers to use in the analysis of bearing chamber film oil flow and heat transfer processes in oil systems.

The numerical simulation of film flows poses a particular engineering challenge due to the high computational cost that would be required to resolve the film when using more conventional multi-phase flow methods, such as the Volume-of-Fluid (VOF) approach [2]. Traditional multiphase flow methods such as VOF require the use of very fine computational grids in order to adequately resolve the film velocity and temperature profiles as well as the film interface and its curvature. This often results in a prohibitively high computational cost for many industrial thin-film flows with a grid size typically one order of magnitude smaller than the minimum film thickness. However, the VOF approach has been shown in numerous studies to be made amenable to the solution of larger scale thick-film flows at a reasonable computational cost.

In order to address this constraint, the Eulerian Thin-Film modelling (ETFM) approach has been presented by a number of authors such as [3, 4, 5, 6] as an alternative. A depth-averaging approach is often adopted, in which the three-dimensional (3D) Navier-Stokes equations are depth-averaged across the film thickness to obtain a set of 2D thin-film equations such as in [3, 4, 5, 6]. The ETFM approach allows for thin-film effects on the core flow and heat transfer to be accounted for without the need to explicitly resolve the very

small scale film flows using a numerical grid as illustrated in Figure 2.

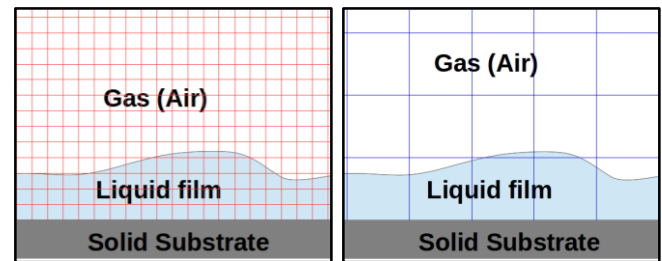


Figure 2: Illustration of film and interface resolution requirements for VOF (left) and ETFM (right) approaches to thin-film modelling

A comparative analysis of the relative performance of the two approaches – VOF and ETFM – for the simulation of thin-film flows over a rotating disk has previously been reported by [7]. The results showed that the ETFM model was able to provide comparable accuracy to VOF at a much lower computational cost. Although the thin-film approach offers considerable advantages in terms of reduced computational cost, there are accuracy and stability benefits in ensuring that in areas where the thin-film approximation is no appropriate valid, a VOF approach is used. This would typically be the case in areas of the bearing chamber, such as the sump regions, where the film velocity field becomes largely three-dimensional and the thin-film approximation breaks down. In these areas, the VOF approach would provide a more accurate prediction of film behaviour.

There is a considerable benefit in developing a coupled ETFM-VOF approach where the depth-averaged ETFM approach is used to simulate the oil flow in very thin-film regions where film thicknesses are below the affordable grid resolution. While elsewhere in the domain, the ETFM model would transition into a Volume-of-Fluid (VOF) approach.

This study presents the formulation of a coupled ETFM-VOF simulation approach implemented in commercial code ANSYS FLUENT v16.0. The resulting coupled ETFM-VOF model has been applied to a 3D shear driven rimming flow test case that is representative of an idealised bearing chamber. Results from the ETFM-VOF simulation are compared against fine grid VOF predictions for the equivalent case.

2. Numerical formulation

2.1. VOF model

The VOF model [2] is a multiphase flow model for the numerical simulation of two or more immiscible fluids with fluid-fluid interface tracking. The method tracks different fluid phase regions within the domain using a volume fraction function given by:

$$\alpha_i = \frac{V_i}{V_T}, \quad (1)$$

where V_i is the volume of phase i within the cell whose total volume is V_T . In any given control volume, the sum of all the volume fractions, α_i , is equal to unity. i.e.

$$\sum_{i=1}^N \alpha_i = 1. \quad (2)$$

In order to obtain the rate of change in any given control volume, a VOF continuity (mass conservation) equation (Equation (3)) is solved for each phase. A single set of momentum equations defined given by Equation (4) is shared by all the phases and used to track the motion of different phases through the domain. The mixture fluid properties (such as density and viscosity) used in computing the shared momentum equation are obtained from a volume-averaging of fluid properties from all the phases present in a control volume as shown in Equations (5) and (6).

$$\frac{\partial \alpha_i}{\partial t} + \frac{\partial}{\partial x_j} (\alpha_i u_j) = S_m, \quad (3)$$

$$\begin{aligned} \frac{\partial}{\partial t} (\rho_m u_i) + \frac{\partial}{\partial x_j} (\rho_m u_i u_j) \\ = -\frac{\partial p}{\partial x_i} + \mu_m \frac{\partial^2 u_i}{\partial x_j \partial x_j} + \rho_m g \\ + S_{i,MOM}, \end{aligned} \quad (4)$$

$$\rho_m = \sum_{i=1}^N (\alpha_i \rho_i), \quad (5)$$

$$\mu_m = \sum_{i=1}^N (\alpha_i \mu_i). \quad (6)$$

The interface between two fluids is located as an iso-surface with a volume fraction in the range $0 < \alpha_i < 1$. A geometric reconstruction scheme is used to reconstruct the interface using a piecewise linear approach. Based on the re-interpolated interface within a given control volume, the advection of fluid through the cell faces may then be computed.

2.2. ETFM model

In the ETFM model, the thin-film flow is idealised as a two-dimensional incompressible Newtonian liquid of density, ρ_l and viscosity, μ_l flowing over a solid substrate and with a free-surface exposed to an incompressible Newtonian gas of ρ_g and viscosity, μ_g .

The film has a spatially varying height, $h(s, t)$ and flows with a film velocity $u(s, y, t)$ - where s is the horizontal flow direction and y is the normal direction as shown in Figure 2. The resulting film flow dynamics over the solid-substrate may

be described by the depth averaged continuity and momentum equations given by Equations (7) and (8).

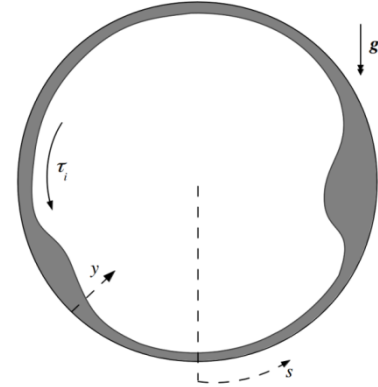


Figure 2 Thin-film rimming flow geometry and coordinate reference system used in the ETFM approach

$$\frac{\partial h}{\partial t} + \frac{\partial q_j}{\partial x_j} = S_m, \quad (7)$$

$$\begin{aligned} \frac{\partial q_i}{\partial t} + \frac{\partial}{\partial x_j} \int_0^h u_i u_j dy = -\frac{h}{\rho_l} \frac{\partial p_f}{\partial x_j} + \frac{h}{\rho_l} \frac{\partial \sigma \kappa_j}{\partial x_j} + g_j h \\ + S_{\tau,j} + S_{MOM}. \end{aligned} \quad (8)$$

Where $p_f = (p_{\text{gas}} - \rho_g g_y h)$, is the film pressure which has a component from the interfacial gas pressure, p_{gas} , and the film hydrostatic pressure, $\rho_g g_y h$. p_f is used to compute the film hydrostatic pressure gradient term, which is the first term on the right hand side (R.H.S.) of Equations (8). S_m and S_{MOM} are generic mass and source terms for the film domain.

Surface tension effects are represented in the surface tension term, $\left(\frac{h}{\rho_l} \frac{\partial \sigma \kappa}{\partial x_j} \right)$, where σ is the surface tension coefficient for the liquid-gas interface, and the interface normal curvature in the j -direction, κ_j , is estimated according to [8]:

$$\kappa_j = \frac{\frac{d^2 h}{dx_j^2}}{\left(1 + \left(\frac{dh}{dx_j} \right)^2 \right)^{1.5}}. \quad (9)$$

The third term on the R.H.S. of Equations (8) represents the momentum source term due to film gravitational body forces in the direction of the film flow. Finally, the fourth source term on the R.H.S. of Equations (8), S_{τ} , represents the balance of viscous shear forces on the film, including contributions from the interfacial shear stress driving the film, τ_i , and the wall shear stress resisting fluid flow over the solid substrate, τ_w . Using appropriate sub-models to estimate τ_i and τ_w , such as in [4], the viscous source term, S_{τ} may be computed according to Equation

$$S_\tau = \frac{\tau_i - \tau_w}{\rho_l} \quad (10)$$

2.3. Coupled ETFM-VOF

VOF and ETFM modelling capabilities available in commercial CFD code ANSYS FLUENT version 16 have been coupled using a User Defined Function (UDF) developed by the authors to facilitate the multi-scale simulation of thin-film flows. The resulting coupled non-isothermal VOF-ETFM formulation is summarised in equations (11) - (12) for the VOF regions and equation (14) - (15) for the ETFM regions with coupling source terms.

$$\frac{\partial \alpha_i}{\partial t} + \frac{\partial}{\partial x_j} (\alpha_i u_j) = S_{m-VOF}, \quad (11)$$

$$\begin{aligned} \frac{\partial}{\partial t} (\rho_m u_i) + \frac{\partial}{\partial x_j} (\rho_m u_i u_j) \\ = -\frac{\partial p}{\partial x_i} + \mu_m \frac{\partial^2 u_i}{\partial x_j \partial x_j} + \rho_m g_j \\ + S_{MOM-VOF,j}, \end{aligned} \quad (12)$$

$$\frac{\partial T}{\partial t} + \frac{\partial}{\partial x_j} (\rho_m T u_j) = \left(\frac{k_m}{\rho_m C_{p,m}} \right) \frac{\partial^2 T}{\partial x_j \partial x_j} + S_{T-VOF,j}, \quad (13)$$

$$\frac{\partial h}{\partial t} + \frac{\partial q_j}{\partial x_j} = S_{m-ETFM}, \quad (14)$$

$$\begin{aligned} \frac{\partial q_i}{\partial t} + \frac{\partial}{\partial x_j} \int_0^h u_i u_j dy = -\frac{h}{\rho_f} \frac{\partial p_f}{\partial x_j} + \frac{h}{\rho_f} \frac{\partial \sigma \kappa_j}{\partial x_j} + g_j h \\ + S_{\tau,j} + S_{MOM-ETFM,j}, \end{aligned} \quad (15)$$

Where S_{m-VOF} and S_{m-ETFM} , are the mass transfer source terms for the VOF domain and the ETFM domain respectively; $S_{MOM-VOF,j}$ and $S_{MOM-ETFM,j}$ are the VOF and ETFM momentum transfer source terms respectively. $S_{T-VOF,j}$, and S_{T-ETFM} are the energy transfer source term for the VOF and ETFM domains respectively. These source terms are jointly used to couple the VOF domain with the ETFM domain and effect a transition from one to another through conservative mass, momentum and energy transfer between the two domains based on a given transition criteria.

The interface between the two models is located at the wall adjacent cells for the VOF domain, yielding volumetric source terms, while for the ETFM domain the transition points are wall faces and source terms are specified as fluxes. The transition from one representation to the other is triggered using local mesh resolution criteria. Figure 3 illustrates this proposed grid resolution based criteria for model selection.

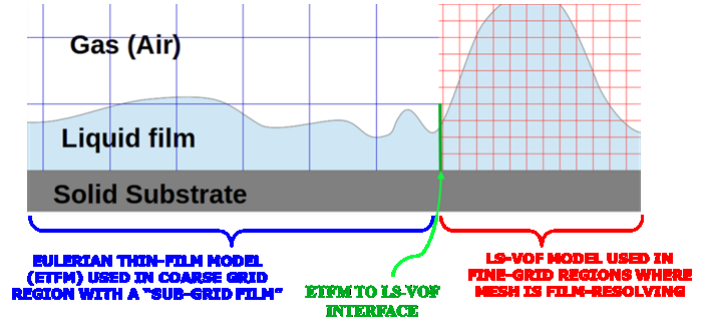


Figure 3: Proposed VOF to ETFM coupling approach

The sufficiency of the local mesh resolution will be assessed based on the total volume fraction ($\alpha_{i,TOTAL}$) of liquid from both ETFM and VOF components that is resident in a given cell,

$$\alpha_{i,TOTAL} = \alpha_{i,VOF} + \alpha_{i,ETFM}. \quad (16)$$

A cut-off volume fraction ($0.5 < \alpha_{i,CRIT} < 1$) of liquid present in the cell from both ETFM and VOF liquid components will be defined and used as measure of the sufficiency of the grid. For cells with a total liquid volume fraction that is above the cut-off value (including those completely filled with liquid), the film-resolving VOF method will be used and all the film mass will be transferred from the ETFM domain to the VOF domain. Conversely, if a cell has a total liquid volume fraction below the cut-off value, mass and momentum from any VOF liquid component will be moved to the ETFM domain. i.e.;

$$\dots \text{if } \begin{cases} \alpha_{i,TOTAL} < \alpha_{i,CRIT}, & \text{switch to ETFM} \\ \alpha_{i,TOTAL} \geq \alpha_{i,CRIT}, & \text{switch to VOF} \end{cases} \quad (17)$$

Subsequently, grid adaptation may be used to ensure sufficient mesh resolution is available for the VOF regions, while keeping the ETFM mesh regions relatively coarse to save on computational cost. The baseline grid would initially be sized based on the expected oil filling fraction (or mean film thickness) such that higher film-thickness regions such as shock/pool regions would result in a switch to the VOF mode. During the transfer between models, global mass and momentum conservation are enforced through the source terms.

3. Simulations

3.1. Setup and cases

The shear driven rimming flow inside an annular chamber has been simulated using both the coupled ETFM-VOF approach and the VOF method.

The domain, illustrated in Figure 4, is composed of an annular space with a stationary outer wall of radius 0.074m and a rotating inner wall of radius 0.064m. The chamber width in the axial direction (z-direction) is 0.015m with the side walls

modelled as stationary wall boundaries. For all cases, the inner shaft is rotating at 7000 RPM.

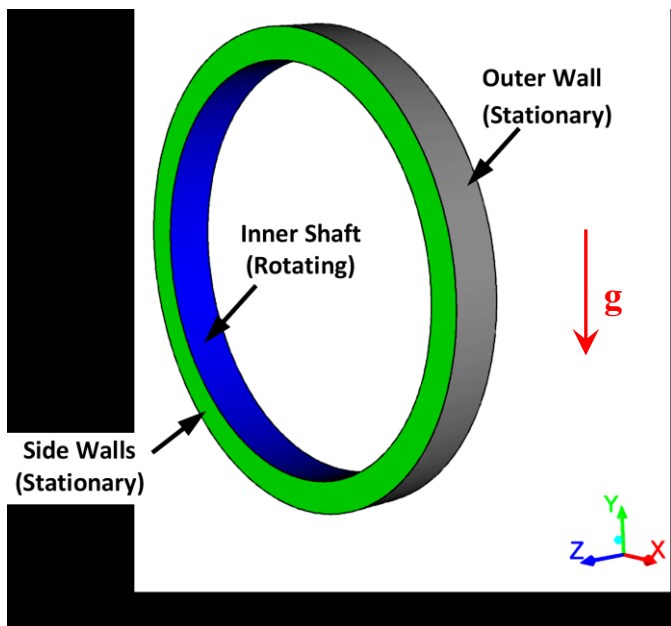


Figure 4: Computational domain and boundaries of test case

The stationary outer wall of the chamber is coated with a liquid film which is driven around the chamber by the air shear induced by the rotating inner shaft. Table 1 summarises the fluid properties used in the simulations. Water properties are used for the liquid while the gas is treated as air and both fluids are assumed to be incompressible and isothermal. The flow is assumed to be turbulent and the k-omega SST turbulence model is used to account for flow turbulence effects.

Table 1: Fluid properties

Fluid	ρ [kg/m ³]	μ [Pa.s]	σ [N/m]
Gas (Air)	1.225	1.7894e-5	0.042
Liquid (Water)	998.2	1.0e-3	

Two structured hexahedral meshes have been used to discretise the domain – a coarse 204,000 cell mesh and a finer 848,000 cell mesh. The mesh details are summarised in Table 2, while Figure 5 illustrated the mesh density from both grids using a cross-section through the annular space.

Table 2: Simulation Cases

Case	Grid	Cells	y+	y _p (m)	Method
1	Coarse	204000	2.92	4E-5	ETFM-VOF
2	Fine	848000	0.25	4E-6	VOF

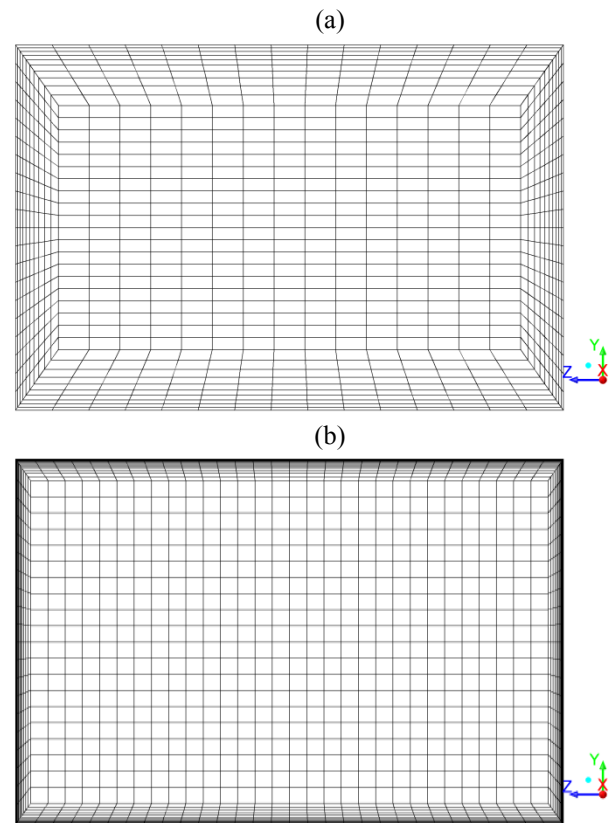


Figure 5: Crosssection through the annular space of; (a) Coarse grid used for the coupled ETFM-VOF solution; (b) Fine grid used for the VOF solution

Initially the simulation is run with air only in order to obtain a steady gas flow velocity induced by the rotation of the shaft. The outer walls of the chamber are coated with a uniform liquid film thickness (h_o) of 0.57 mm and transient CFD simulations run until a stationary state is reached corresponding to selected monitors of velocity oscillating about a stable mean as shown in Figure 6.

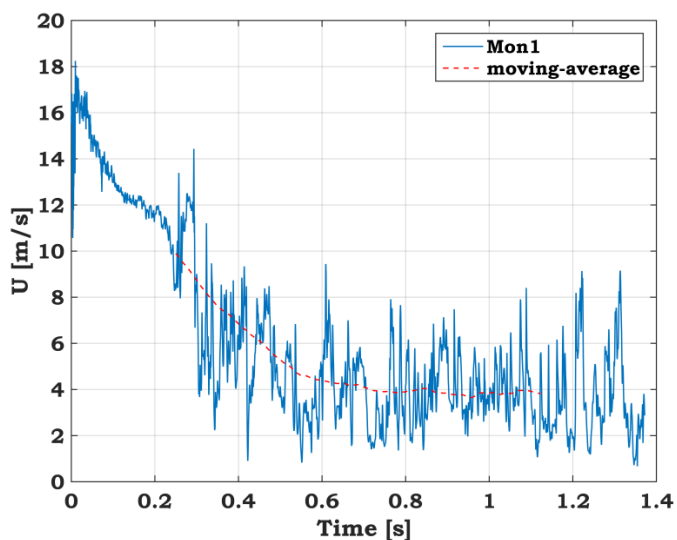


Figure 6: Sample velocity monitor time-series for the coupled ETFM-VOF solution

Two cases are presented in this paper. In the first case, a coupled ETFM-VOF solution has been computed on the coarse grid, while in the second case, a VOF approach is used with a fine grid. With an initial liquid film thickness of 0.57 mm, the coarse grid gives a mesh resolution of 5 cells across the film thickness while the fine grid gives a mesh resolution of 15 cells across the film thickness. The resulting gas phase flow fields and the liquid distribution along the wall are compared between ETFM-VOF and VOF methods as well as their relative computational costs.

3.2. Results

In Case 1, the coarse grid is used to simulate the rimming flow using a coupled ETFM-VOF approach. Contours of the total liquid volume fraction, $\alpha_{i,TOTAL}$, as defined in Equation (16) are shown in Figure 7 for the stationary outer wall. The liquid forms a pool on the side of the chamber where gravity opposed film motion resulting in a high VOF region while a low VOF region is formed on the opposite side of the chamber.

A critical volume fraction of 0.5 was used to trigger the transition in these simulations; as a result, only regions along the wall with $\alpha_{i,TOTAL} > 0.5$ retain a VOF representation, with all other regions having the local liquid mass and momentum transferred into the ETFM solution in the adjacent wall face. This is illustrated by the contours of VOF liquid volume fraction as shown in Figure 8 where all regions previously shown in Figure 7 to have a total volume fraction of less than 0.5 are shown to have zero liquid volume fraction in the VOF solution. The corresponding film thickness from the ETFM model in these regions is shown in Figure 9.

The images in Figure 7 - Figure 10 are instantaneous snapshots showing the representative film distribution along the wall for the stationary solution. During the simulation, some cells will transition from ETFM to VOF representations and vice-versa depending on the local volume fraction of liquid in the cell.

The ETFM-VOF simulations were run on a cluster with Intel Xeon E5-2670 processors and required approximately 6,650 CPU Hours for each second of flow simulated.

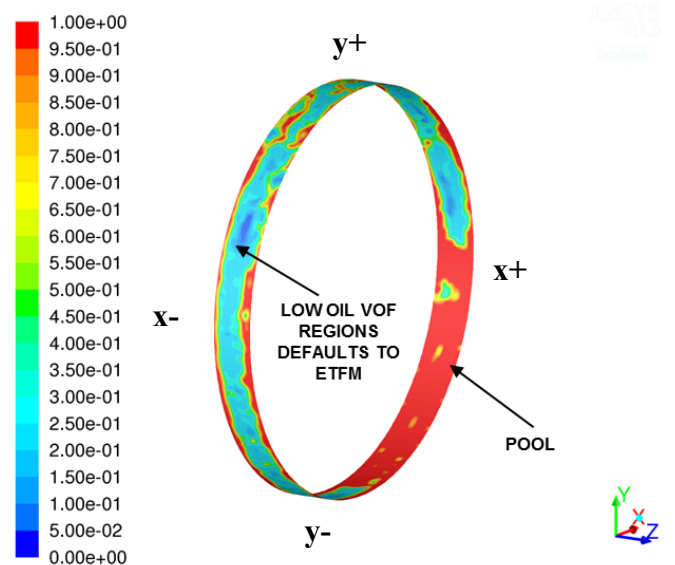


Figure 7: Contours of total liquid volume fraction ($\alpha_{i,TOTAL}$) for the coupled ETFM-VOF simulation (case 1)

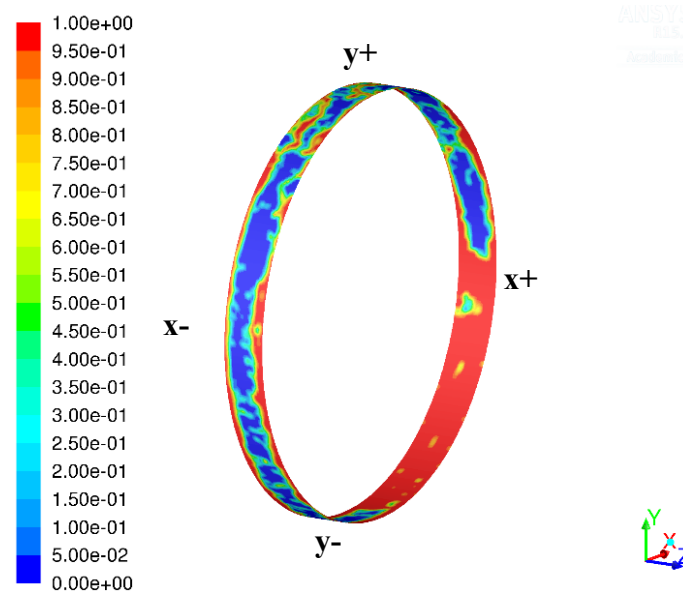


Figure 8: Contours of VOF liquid volume fraction ($\alpha_{i,VOF}$) for the coupled ETFM-VOF simulation (case 1). $\alpha_{i,VOF} = 0$ in ETFM regions

In Case 2, a VOF solution of the same test case was obtained using the fine mesh described in Section 3.1. As shown in Figure 11, the higher grid resolution in the VOF solution predicted similar features to the ETFM-VOF solution on the coarser grid (Figure 7) such as the pool formation on the right-

hand-side of the domain and the low volume fraction region on the left hand side of the domain.

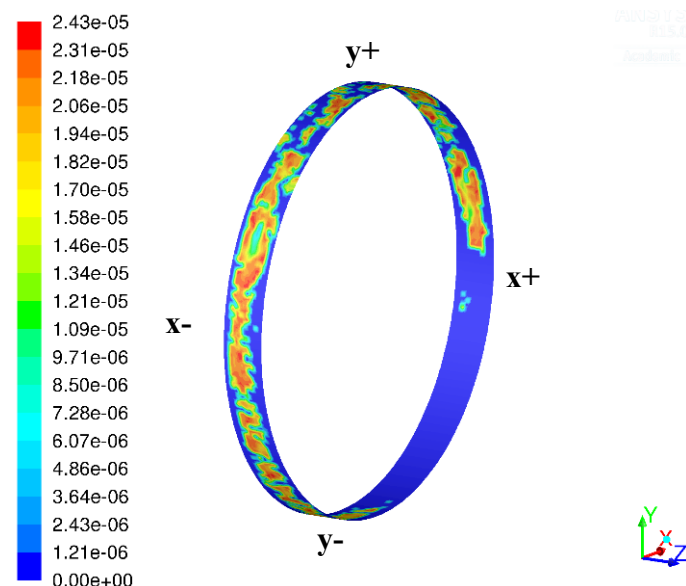


Figure 9: Contours of film thickness (m) in the ETFM region for the coupled ETFM-VOF simulation (case 1)

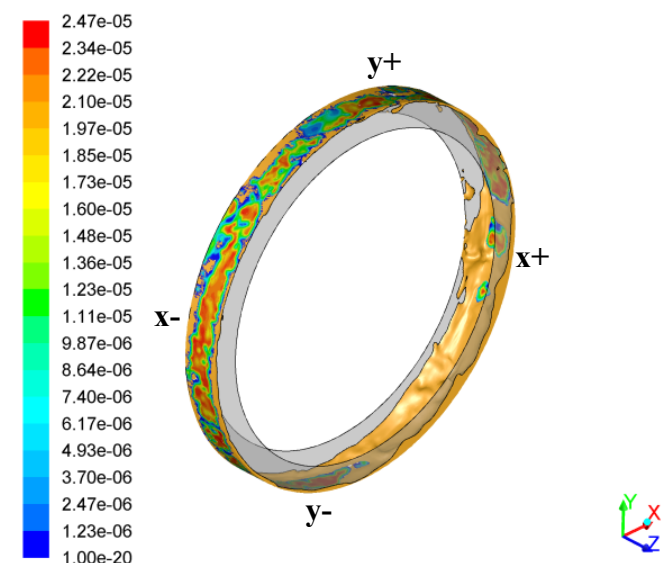


Figure 10: Contours of film thickness (m) in the ETFM region of the stationary outer wall, together with iso-surfaces of $\alpha_{i,VOF} = 0.5$ showing the VOF gas-liquid interface in the coupled ETFM-VOF simulation (case 1)

The ETFM-VOF model predicted no dry patches on the wall, but a persistent thin-film in the low volume fraction region. However, the VOF solution predicted film dry-out in some locations as shown in Figure 11. This dry-out, with patches of $\alpha_{i,VOF} = 0$ is a result of the presence of very low liquid volume fraction regions below the prescribed VOF cut-off

value of $\alpha_{min,VOF} = 1E-6$ for ANSYS FLUENT [8]. Values below this cut-off are set to zero resulting in film dry-out. An even finer grid would be required in order to resolve the film thickness in these regions with a traditional VOF simulation. For the coupled ETFM-VOF solution (Figure 10), the film thickness and flux in these regions may be estimated using an even coarser grid than that used for the VOF simulation.

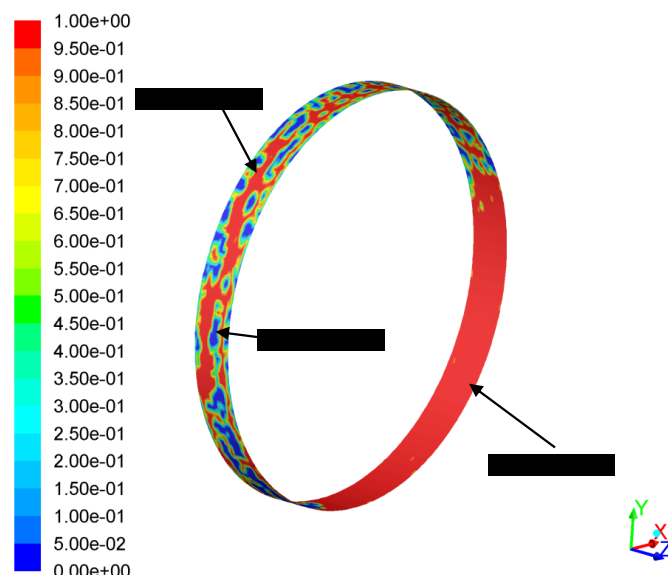


Figure 11: Contours of VOF liquid volume fraction ($\alpha_{i,VOF}$) for the VOF simulation (case 2). $\alpha_{i,VOF} = 0$ represents drying ETFM regions

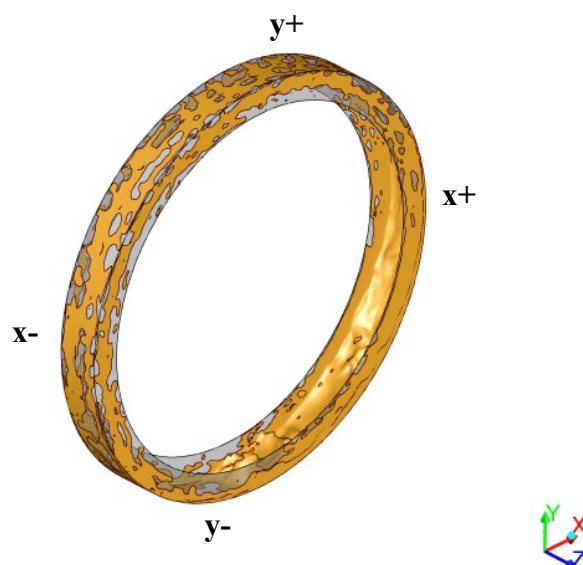


Figure 12: Iso-surfaces of $\alpha_{i,VOF} = 0.5$ showing the VOF gas-liquid interface in the VOF simulation (case 2)

The VOF simulations were run on an HPC cluster and required approximately 22,700 CPU Hours for each second of flow time simulated compared to 6650 CPU Hours for each second of the ETFM-VOF case simulated. This relatively higher computational cost is mainly the result of the increased grid

resolution requirements for the VOF simulation compared to the ETFM-VOF solution.

A vertical section through the domain showing the representative film distribution is shown in Figure 13 for both the VOF case and the ETFM-VOF case. Both cases predict the formation of a liquid pool although the depth and start location of the pool differs between the two cases.

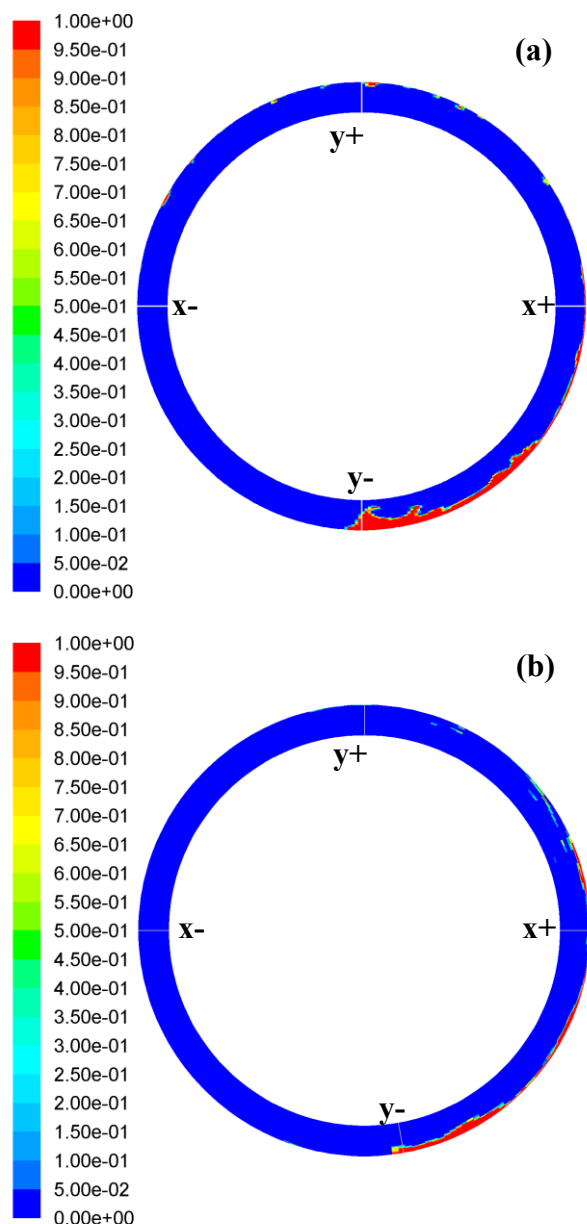


Figure 13: Contours of VOF volume fraction (cell centre values) on a vertical section through the centre of the domain showing the film distribution in; (a) the fine grid VOF case (case 2); (b) the coupled ETFM-VOF case (case 1)

Velocity and film thickness profiles in the radial direction are extracted at line locations shown in Figure 13 in order to contrast the two methods. These profiles are shown in Figure 14 for the cell phase distribution and Figure 15 for the velocity profiles. In both cases, the plots of phase distribution show that

while sufficient grid resolution exists to resolve the flow in the thick film or pool region (line y-), in the thin film region such as line x- (shown scaled up in-set in Figure 14) where a liquid volume fraction of 1E-5 is predicted by the VOF solution in the first cell along the wall, there is insufficient grid resolution and in some cases the volume fraction in these cells would fall below the VOF cut-off resulting in an unphysical film dry-out prediction for traditional VOF methods.

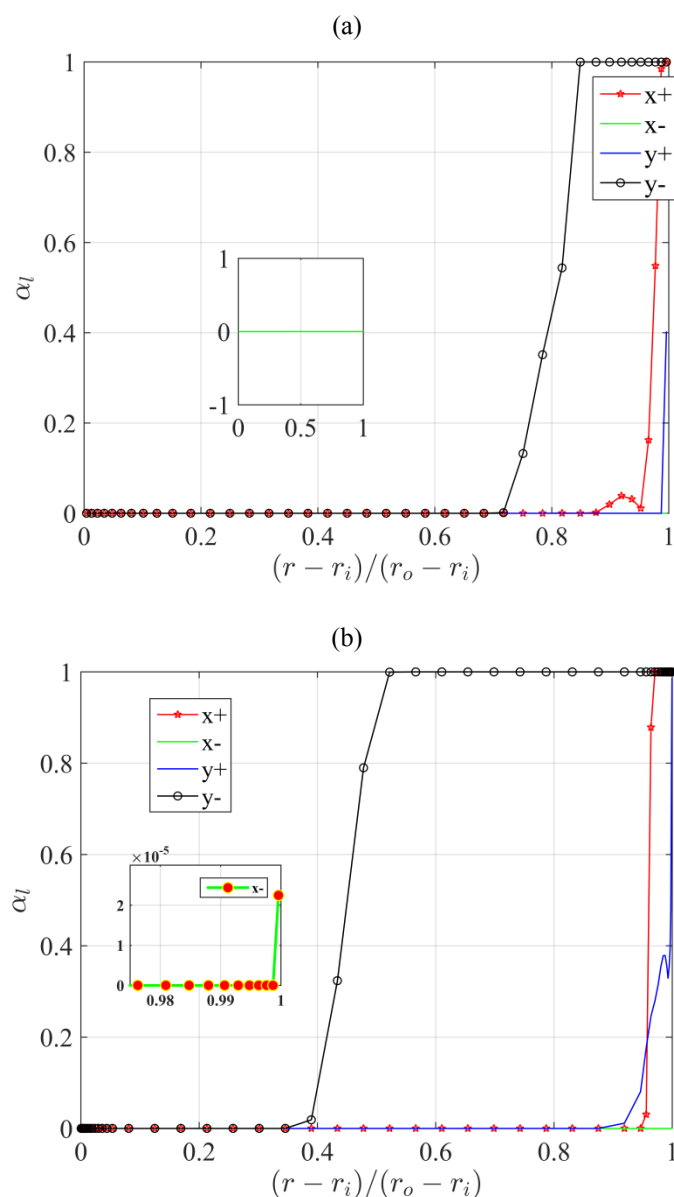


Figure 14: Radial profiles of VOF liquid volume fraction across the chamber in (a) the coupled ETFM-VOF case (case 1) and (b) the fine grid VOF case (case 2). r_i is the inner shaft radius, r_o is the outer wall radius and line positions (x+, x-, y+, y-) are shown in Figure 13. Inset is a scaled-up plot for position x- showing the VOF liquid volume fraction for the x- position.

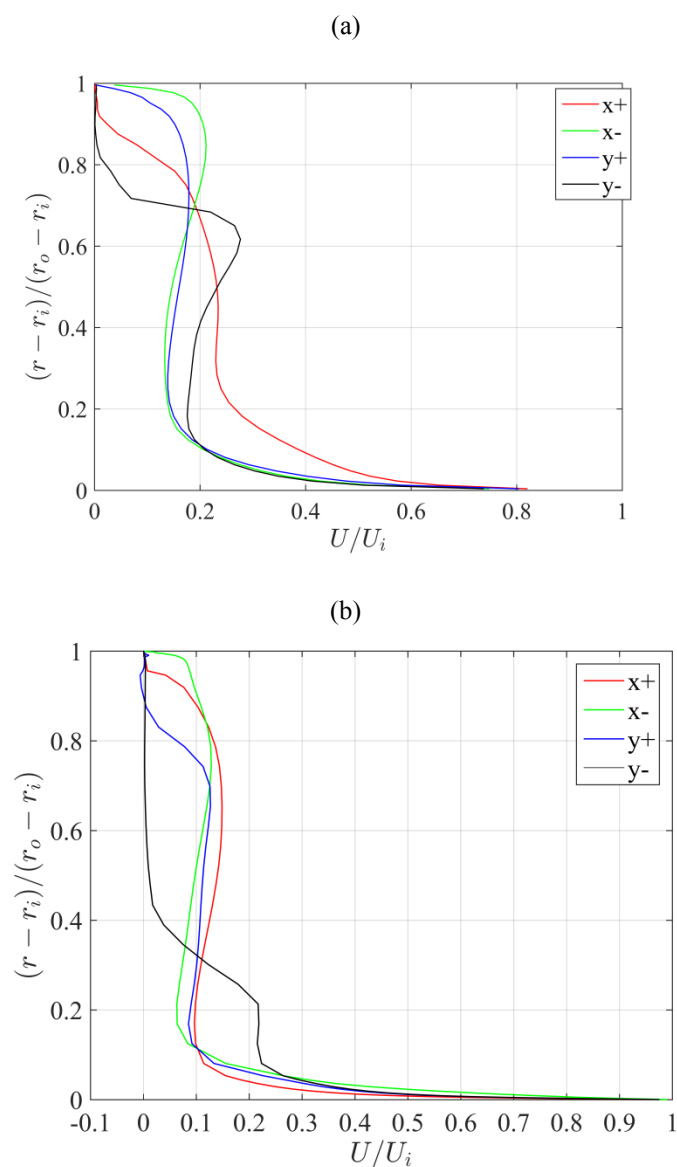


Figure 15: Radial profiles of normalised velocity along lines across the chamber for; (a) coupled ETFM-VOF(case 1); (b)VOF (Case 2). U_i is the inner shaft speed, r_i is the inner shaft radius, r_o is the outer wall radius and line positions are shown in Figure 13

The ETFM-VOF film velocity profiles show quantitative differences from the VOF solution and this may be largely attributed to the quantitative differences in film distribution, with the VOF having thicker films. A possible source of these differences is the effect of turbulence damping at the liquid-gas interface in VOF on the VOF film distribution. In the ETFM model, the film wall is treated as a film wall and turbulence damping is carried out; by default additional source terms are required to accurately capture this turbulence damping in a VOF simulation. Further investigations into the role of turbulence damping on the ETFM-VOF solution are ongoing. The ETFM-VOF approach is demonstrated to be feasible and to produce stable solutions with film profiles and phase

distributions that are qualitatively similar to the more computationally expensive VOF solutions.

Additional validation studies are required in order to assess the relative accuracy of the two approaches as well as the effects of grid sensitivity and selection of values for critical volume fraction, $\alpha_{i,CRIT}$.

4. Conclusions

A coupled ETFM-VOF formulation for the simulation of multi-scale film flows has been presented. A rimming flow test case is used to demonstrate the method and test comparisons are made with a high resolution VOF solution. Comparable predictions are provided between the ETFM-VOF and the fine grid VOF solutions with both approaches predicting the formation of a liquid pool in the side of the chamber where gravity opposed film motion and a very thin-film region on the opposite side where gravity acts in the same direction as the film.

The ETFM-VOF solution is shown to provide a considerable saving in computational costs due to the relaxed grid resolution requirements. In addition, no artificial dry-out of the film is predicted as even very low liquid volume fractions may be accounted for using the ETFM approach where they are below the VOF threshold.

Further studies into the role of interface turbulence damping models in the VOF solution on the coupled ETFM-VOF solution are suggested to evaluate some of the discrepancies observed between the fine-grid VOF solution and the coupled ETFM-VOF solution. This paper presents a preliminary investigation and further investigations are required to validate the coupled ETFM-VOF approach against experimental data.

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